

The **Sigmoid activation function** is commonly used in neural networks when we want to convert a value into a number between **0 and 1**.

**e≈2.718** is a special mathematical constant called **Euler's number**.  
It is a constant

Z = In put value

| Z     | After Sigmoid |
|-------|---------------|
| -5.50 | 0.00          |
| -4.60 | 0.01          |
| -4.00 | 0.02          |
| -3.40 | 0.03          |
| -2.80 | 0.06          |
| -2.20 | 0.10          |
| -1.60 | 0.17          |
| -1.00 | 0.27          |
| -0.40 | 0.40          |
| 0.20  | 0.55          |
| 0.80  | 0.69          |
| 1.40  | 0.80          |
| 2.00  | 0.88          |
| 2.60  | 0.93          |
| 3.20  | 0.96          |
| 3.80  | 0.98          |
| 4.40  | 0.99          |
| 5.00  | 0.99          |
| 6.50  | 1.00          |
| 8.00  | 1.00          |

$$SigmoidActivation = \frac{1}{1 + e^{-z}}$$

$e = 2.718$

$$= \frac{1}{1 + 2.718^{-(-5.50)}}$$

$= 0.00$

$$= \frac{1}{1 + 2.718^{-(-4.60)}}$$

$= 0.01$

$$= \frac{1}{1 + 2.718^{-(-4.00)}}$$

$= 0.02$